Quick Introduction:

The user study will have DTAG's ASGA system as a use case to identify explainability requirements for AI models. The ASGA system, as of today, does not integrate an AI-based approach, but it is under investigation on how AI-based support can improve the system's behavior.
Please focus on Req1.1 to Req1.6. The requirements before and after provide context.

Active Side Guard Assist (ASGA) is an advanced driver assistance system (ADAS) designed to prevent side collisions—particularly during turning maneuvers in urban environments—by monitoring the vehicle's blind spot and responding to nearby vulnerable road users (VRUs), such as pedestrians and cyclists. With the growing need for safer urban mobility and autonomous systems, integrating Artificial Intelligence (AI) into ASGA systems has become a critical enabler for more intelligent, adaptive, and reliable safety interventions.
At the core of modern ASGA systems lies a multi-modal AI pipeline that leverages deep learning and sensor fusion to perceive, understand, and respond to dynamic side-environment scenarios in real time. AI enables ASGA to go beyond traditional rule-based detection by introducing context-aware understanding, trajectory forecasting, and adaptive decision-making.
The typical AI-powered ASGA architecture consists of three major components:

1. Perception Module: Utilizes convolutional neural networks (CNNs) or transformer-based vision models to detect and classify nearby objects from side-facing cameras. Sensor fusion techniques combine visual data with inputs from radar, LiDAR, and ultrasonic sensors to enhance robustness and reduce ambiguity in complex environments.

2. Prediction and Risk Assessment Module: Predicts the motion trajectories of detected objects using temporal AI models (e.g., LSTMs, attention-based models), estimating the likelihood of collision during vehicle maneuvers like turning or lane changes. AI enhances the system's ability to distinguish between static and dynamic threats.

3. Decision and Intervention Module: Based on real-time risk assessment, the system determines the appropriate action—ranging from driver alerts to automatic braking. Reinforcement learning or expert systems can be employed to optimize intervention policies under varying urban driving conditions.

By harnessing AI, ASGA systems become more adaptive to diverse road contexts, capable of generalizing across unseen scenarios, and more accurate in reducing false positives and negatives, ultimately contributing to the vision of zero traffic fatalities and fully autonomous urban mobility.

Side Guard Assist (SGA) - GSR regulatory:

- Information: The SGA system intends to fulfill and extend the UN ECE R151 BSIS, which shall support drivers in monitoring objects on the passenger side where visibility is limited.
- Req1: The SGA shall monitor the tractor and trailor subzones for objects in state moving or moving_stopped.

---------- AI requirements ----------

- Req1.1: The AI model shall accurately identify and classify objects within the monitored subzones, including pedestrians, cyclists, and other vehicles.
- Req1.2: The AI model shall differentiate between stationary and moving objects to provide relevant warnings based on object behavior.
- Req1.3: The AI model shall operate effectively under various environmental conditions, such as rain, fog, and low light.
- Req1.4: The AI model shall process data from multiple sensors (see Req4) to enhance object detection accuracy.
- Req1.5: The AI model shall continuously learn and improve its object detection capabilities through real-time data analysis and updates.
- Req1.6: The AI model shall minimize false positives and false negatives to ensure reliable warnings and interventions.

---------- AI requirements ----------

- Req2: The SGA shall warn the driver in case of a risk for "collision with vulnerable road users during turning" and "collision during lane change".
- Req3: The SGA shall warn the driver by the two states "information" (yellow visual warning) and "warning" (red visual warning && warning tone).
    - The yellow "information" level gives the driver the chance to check the immediate surroundings.
    - The red "warning" level is designed to prompt the driver to respond/alter his/her driving behavior.
- Req4: The SGA detection shall be realized with the support of
    - 4 short range radars (2 x left and 2 x right),
    - 1 multi purpose camer (front) and
    - 1 long range radar (front).

Active Side Guard Assist (ASGA1):

- Information: ASGA1 intends to avoid collisions during turning.
- Req5: The ASGA1 shall extend the SGA by a third state named "brake application active" (red visual warning && warning tone).
    - The brake application is executed when there is an immediate risk for collision with an object classified as moving.
- Req6: The ASGA1 shall initiate braking if the SGA has generated a warning level of 2 for the same object.
- Req7: The ASGA1 shall initiate braking up to a maximum vehicle velocity of 20 kph.

Active Side Guard Assist (ASGA2):

- Information: ASGA2 intends to avoid collisions during lane change.
- Req8: The ASGA2 shall initiate steering intervention if
    - Turn indicator in direction of lane change active
    - Vehicle is moving away from middle of own lane towards object
    - Ego speed equal or higher 60 kph
    - Lane data valid
    - Safety checks for lane data valid
    - Width of lane plausible
    - Red warning on moving object active
    - Steering function parameter check is valid

Information on object detection classes

- Information: The below object classes exist nowadays for the object detection of the ASGA system. It shall provide further context.

Due to Company Privacy Policiy and confidentiality, the following information has been omitted.